

TRIZ Principles

As per Creative Thinking and Innovation course syllabus

Syllabus

Common features of good solutions – resolve contradiction, use available resource, increase the ideality, trade-off, inherent contradiction,

30 key TRIZ principles – multifunction, preliminary action, compensation, nested doll, blessing in disguise, segmentation, separation, regional influences, symmetry change, opaque & porous, inflate and deflate, color, recycle & recover, phase transformation, energy, imaging, environment, composition, economical, surface response, equipotential, static & dynamic, continuous & intermittent, servo systems, smart systems, dimensions

Please refer to the handouts provided earlier for the first part. Also the detailed explanations for each of the principles can be found in those handouts. This handout is only for quick reference.

Multifunction

Eg. your very own digital multimeter

It has multiple functionalities put in one device

Like dc and ac voltage / current

Resistance across wide ranges

Component and continuity tester

Advantages: Reduced cost, convenient to use

Disadvantages: failure leads to higher cost



Multifunction



Remembering: What is the definition of the multifunction principle in TRIZ?

Answer: The multifunction principle in TRIZ refers to a design approach where a single component or system performs multiple functions, thereby increasing efficiency and reducing complexity.

Understanding: How does the multifunction principle enhance product design?

Answer: The multifunction principle enhances product design by integrating several capabilities into one device, which can lead to reduced material usage, lower production costs, and improved user convenience.

Applying: Provide an example of a product that effectively utilizes the multifunction principle.

Answer: A *Swiss Army knife* is a classic example, as it combines multiple tools (knife, screwdriver, scissors) into one compact device, serving various functions for users.

Multifunction



Analyzing: Analyze how multifunctionality can lead to cost savings in product manufacturing.

Answer: By reducing the number of components needed in a product, multifunctionality can lower material costs, decrease assembly time, and minimize inventory requirements, ultimately leading to significant cost savings.

Evaluating: Evaluate the advantages and disadvantages of implementing multifunctional designs in consumer products.

Answer: Advantages include space-saving designs and increased convenience for users. Disadvantages may involve complexity in use and potential for higher failure rates if one function fails.

Preliminary Action



Remembering: What does the preliminary action principle entail in TRIZ?

Answer: The preliminary action principle involves taking proactive steps to prevent problems before they occur in a process or system.

Understanding: Explain how preliminary actions can prevent problems in engineering processes.

Answer: By anticipating potential issues and implementing preventive measures, such as quality checks or risk assessments, engineers can avoid costly delays and failures in projects.

Preliminary Action



Applying: Give an example of a preliminary action taken in a project to mitigate risks.

Answer: In software development, conducting thorough testing before deployment is a preliminary action that helps identify bugs and reduce post-launch issues.

Analyzing: Analyze a case where a lack of preliminary action led to project failure.

Answer: The Boeing 737 MAX incidents illustrate how insufficient preliminary safety checks led to catastrophic failures and significant financial losses for the company.

Evaluating: Assess the effectiveness of preliminary actions in improving safety in industrial operations.

Answer: Implementing safety training sessions as a preliminary action significantly reduces workplace accidents and enhances overall safety culture within industrial operations.

Compensation



Remembering: What is meant by the compensation principle in TRIZ?

Answer: The compensation principle refers to adjusting one aspect of a system to counterbalance or compensate for changes or deficiencies elsewhere.

Understanding: Describe how compensation can improve system performance under varying conditions.

Answer: Compensation mechanisms can dynamically adjust parameters (like speed or load) to maintain optimal performance despite external changes such as temperature fluctuations or load variations.

Compensation



Applying: Illustrate how compensation can be applied to enhance the durability of a mechanical device.

Answer: Using shock absorbers in vehicles compensates for road irregularities, enhancing ride comfort and protecting sensitive components from damage.

Analyzing: Analyze a situation where compensation was necessary to resolve a design flaw.

Answer: In bridge design, adding tension cables compensates for structural weaknesses due to load-bearing issues, ensuring stability and safety.

Evaluating: Evaluate the role of compensation in balancing trade-offs during product design.

Answer: Compensation allows designers to balance conflicting requirements (e.g., cost vs. quality) by adjusting features without compromising overall functionality.

Nested Doll



This principle emphasizes the arrangement of components within a system to maximize space efficiency and compactness.

Remembering

Question: What is the Nested Doll Principle in TRIZ?

Answer: The Nested Doll Principle involves placing one object inside another to enhance compactness and efficiency within a system, similar to Russian nesting dolls

Nested Doll



Understanding: Explain how the nested doll principle can be applied to improve packaging design.

Answer: Using nested packaging allows products to be shipped more compactly, reducing shipping costs and environmental impact while maximizing storage space.

Applying: Provide an example of how nested structures can enhance functionality in consumer products.

Answer: A set of measuring cups that nest within each other saves kitchen space while providing multiple sizes for cooking tasks.

Nested Doll



Analyzing: Analyze the benefits and challenges of using nested designs in construction materials.

Answer: Benefits include reduced material waste and improved thermal insulation; challenges may involve increased complexity in assembly and structural integrity concerns.

Evaluating: What are the advantages and disadvantages of using the Nested Doll Principle?

Answer: Advantages include improved space utilization, reduced material costs, and enhanced system functionality. Disadvantages may involve increased complexity in design and potential challenges in accessing nested components for maintenance.

Blessing in Disguise



This principle suggests that harmful factors or negative effects can be leveraged to create positive outcomes. Like turning 'lemons into lemonade'.

Remembering: What is meant by "blessing in disguise" in TRIZ?

Answer: A "blessing in disguise" refers to an initial problem or setback that ultimately leads to unexpected positive outcomes or innovations.

Understanding: Describe a scenario where an initial problem led to an unexpected beneficial outcome.

Answer: The development of penicillin was initially accidental; Alexander Fleming's contamination of petri dishes led to discovering its antibacterial properties, revolutionizing medicine.

Blessing in Disguise

Applying: Illustrate how recognizing a blessing in disguise can lead to innovative solutions.

Answer: Companies that pivoted during economic downturns often found new markets or products; for example, distilleries producing hand sanitizer during shortages showcased adaptability leading to new revenue streams.

Analyzing: Analyze a case study where a challenge turned into an opportunity through creative problem-solving.

Answer: Netflix's transition from DVD rentals to streaming services arose from declining DVD sales but became an opportunity that transformed entertainment consumption globally.



Blessing in Disguise

Evaluating

Question: What are the potential advantages and disadvantages of applying the Blessing in Disguise Principle?

Answer: Advantages include fostering creativity, promoting resourcefulness, and encouraging unconventional solutions that can lead to breakthroughs.

Disadvantages may involve risks if harmful factors are not adequately controlled or understood, potentially leading to unintended consequences.



Segmentation



This principle suggests dividing a system or object into smaller, manageable parts to simplify tasks and enhance functionality.

Remembering: What is the segmentation principle in TRIZ?

Answer: The segmentation principle involves breaking down complex systems into smaller, manageable parts or segments that can be individually optimized or modified.

Segmentation



Understanding:* Explain how segmentation can improve assembly and disassembly processes.

* Answer:* Segmentation allows components to be easily assembled or disassembled without affecting other parts, simplifying maintenance and repair processes significantly.

Applying:* Provide an example of segmentation used effectively in product design or manufacturing.

* Answer:* Modular smartphones allow users to replace individual components (like batteries or cameras) instead of replacing the entire device, promoting sustainability and ease of repair.

Segmentation



Analyzing:* Analyze how segmentation can affect cost and efficiency in production lines.

* Answer:* Segmentation enables parallel processing; different segments can be manufactured simultaneously on separate lines, reducing overall production time and costs while increasing throughput.

Evaluating:* Evaluate the trade-offs involved in applying segmentation to complex systems.

* Answer:* While segmentation enhances flexibility and maintenance ease, it may increase initial design complexity and require more precise coordination between segments during assembly.

Separation

Traffic lights are used to sequence the flow of traffic at different points of time.

Reference: TETRUS



The Separation Principles help resolve contradictions by separating opposite requirements in different ways:

Separation in time: Schedule system operations so that contradictory functions take effect at different times.

Separation in space: Partition the system into subsystems and assign contradictory functions to different subsystems.

Separation between part and whole: Partition the system and assign contradictory functions to subsystems.

Separation between conditions: Change the system or environment so that only the helpful process can take place.

Separation

Traffic lights are used to sequence the flow of traffic at different points of time.

Reference: TETRUS



Remembering:* What does the separation principle involve according to TRIZ?

* Answer:* The separation principle involves isolating conflicting elements within a system so they do not interfere with each other's functions or performance.

Understanding:* Describe how separation can resolve conflicts between competing design requirements.

* Answer:* By separating functions (e.g., thermal insulation vs. structural support), designers can create solutions that optimize each requirement without compromising overall performance.

Separation

Traffic lights are used to sequence the flow of traffic at different points of time.

Reference: TETRUS



Applying:* Illustrate an application of separation that improved system performance or safety.

* Answer:* In automotive design, separating fuel tanks from passenger compartments enhances safety by preventing fire hazards during collisions while maintaining vehicle integrity.

Analyzing:* Analyze how separation techniques can be used in material science for better results.

* Answer:* Separating materials based on density during recycling processes improves recovery rates by allowing different materials (like plastics) to be processed independently for better quality outputs.

Evaluating:* Assess the effectiveness of separation methods in reducing manufacturing waste.

* Answer:* Implementing separation techniques such as filtration or centrifugation during production processes significantly reduces waste by ensuring only usable materials are processed further.

Regional Influences



The Regional Influences Principle suggests that the characteristics of a system can be enhanced by considering the influence of its environment or local conditions. This principle encourages adapting designs to suit regional needs and leveraging local resources or conditions to improve functionality and efficiency.

Regional Influences

Remembering:* What are regional influences as defined by TRIZ?

* Answer:* Regional influences refer to local factors such as culture, climate, regulations, and market needs that affect product design and innovation strategies.

Understanding:* Explain how regional factors can impact product design and innovation strategies.

* Answer:* Regional factors influence user preferences; for instance, climate affects material choices (e.g., waterproof materials in rainy regions), while cultural norms shape aesthetics.



Regional Influences



Applying:*

Provide an example of how regional influences have shaped technology development in a specific area.

* Answer:* In Japan, earthquake-resistant technologies have been developed extensively due to frequent seismic activity influencing building designs significantly.

Analyzing:* Analyze the effects of regional influences on global supply chains and manufacturing practices.

* Answer:* Regional influences such as labor costs affect sourcing decisions; companies may relocate production facilities closer to markets with lower labor costs while considering local regulations.

Evaluating:* Evaluate how understanding regional influences can lead to better market penetration strategies for new products.

* Answer:* Companies that adapt their products based on regional preferences often achieve higher acceptance rates; Coca-Cola's localized flavors demonstrate successful market penetration strategies.

Symmetry Change



The Symmetry Change Principle suggests modifying the symmetry of an object or system to enhance its performance or functionality. This principle can involve introducing asymmetry to improve efficiency, effectiveness, or adaptability.

Symmetry Change



Remembering*: What does symmetry change refer to within TRIZ principles?

* Answer*: Symmetry change involves altering symmetrical properties within designs (e.g., shapes) which may enhance functionality or aesthetics.

Understanding*: Explain how changing symmetry can affect functionality or aesthetics in design.

* Answer*: Changing symmetry might improve aerodynamics; asymmetrical wings on aircraft enhance lift while maintaining stability compared with traditional symmetrical designs.

Symmetry Change



Applying*:

Provide an example where symmetry change was successfully applied to solve a design problem.

* Answer*: Asymmetrical bicycle frames improve rider comfort by distributing weight more evenly across different riding positions compared with traditional symmetrical frames.

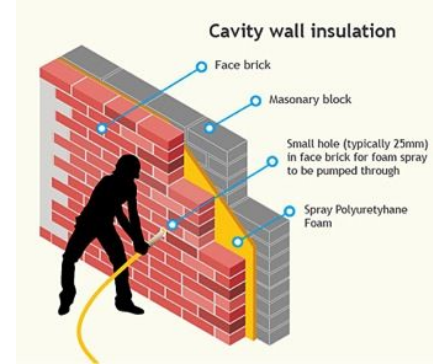
Analyzing*: Analyze how symmetry changes can impact user interaction with products or systems.

* Answer*: Asymmetrical designs may offer unique ergonomic benefits; e.g., asymmetrical handles on tools provide better grip depending on user preference compared with symmetrical handles.

Evaluating*: What are the advantages and disadvantages of applying the Symmetry Change Principle?

Answer: Advantages include enhanced adaptability, improved performance under specific conditions, and increased efficiency. Disadvantages may involve increased complexity in design and potential challenges in manufacturing asymmetrical components.

Opaque & Porous

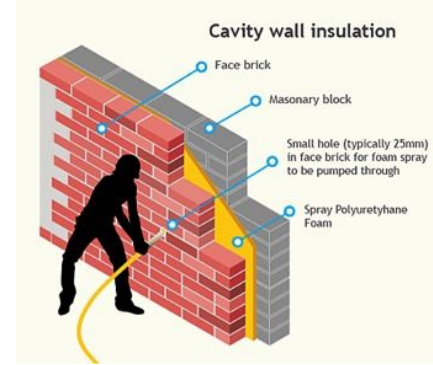


The Opaque & Porous Principle involves utilizing the properties of materials that can either block or allow certain substances to pass through. This principle can be applied to enhance functionality, improve efficiency, or create new interactions within a system.

Remembering: What are opaque and porous materials, and what roles do they play according to TRIZ?

Answer: Opaque materials do not allow light to pass through, while porous materials have tiny holes that allow fluids or gases to pass. In TRIZ, they are used to solve problems related to insulation, filtration, and packaging.

Opaque & Porous



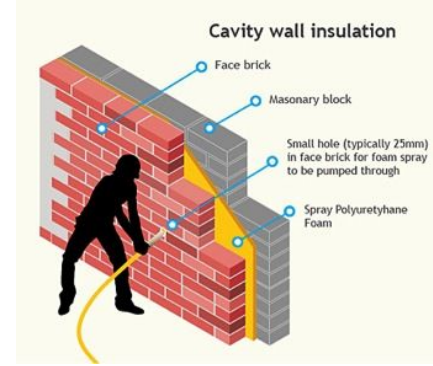
Understanding: Describe how combining opaque and porous materials can create innovative solutions in various applications.

Answer: Combining opaque insulation with porous membranes can create energy-efficient building materials that prevent heat loss while allowing moisture control, enhancing durability and comfort.

Applying: Illustrate an example where opaque and porous materials were used effectively together in design or engineering.

Answer: In soundproofing applications, porous acoustic panels covered with opaque fabric reduce sound transmission while maintaining aesthetic appeal in interior design.

Opaque & Porous



Analyzing: Analyze the benefits and drawbacks of using opaque versus porous materials in specific contexts, such as packaging or construction.

Answer: Opaque materials protect contents from light degradation but limit visibility, while porous materials allow breathability but may compromise protection against external elements. The choice depends on the specific requirements of the application.

Evaluating: Assess how the properties of opaque and porous materials influence their suitability for different applications.

Answer: The suitability of opaque and porous materials depends on their properties; for instance, opaque materials are ideal for shielding sensitive items from light, while porous materials are better for applications requiring ventilation or filtration.

Inflate and Deflate



The Inflate and Deflate Principle involves using inflation or deflation to change the properties or functionalities of an object or system. This principle can be applied to create adaptable structures, enhance performance, or facilitate temporary changes in state.

Remembering: What does the inflate and deflate principle signify within TRIZ?

Answer: The inflate and deflate principle refers to mechanisms that allow components to expand or contract dynamically based on operational requirements, enhancing adaptability in various applications.

Inflate and Deflate



Understanding: Explain how inflating or deflating components can provide advantages in various systems, such as transportation or packaging.

Answer: Inflatable structures can be easily transported when deflated, saving space and weight. When inflated, they provide structural support or cushioning, making them ideal for applications like air mattresses or inflatable boats.

Applying: Provide an example where inflation/deflation mechanisms improved product performance or adaptability.

Answer: Inflatable safety cushions in vehicles deploy during a collision to absorb impact forces, enhancing passenger safety while remaining compact when not in use.

Inflate and Deflate



Analyzing: Analyze situations where inflate/deflate systems have been beneficial or problematic within engineering applications.

Answer: Inflatable structures are beneficial in aerospace for lightweight designs but can be problematic if punctured. Ensuring reliability is crucial for safety-critical applications.

Evaluating: Evaluate the potential risks associated with using inflatable structures compared to rigid alternatives.

Answer: Inflatable structures may offer weight savings but pose risks such as puncture vulnerability and potential failure under extreme conditions compared to rigid structures that provide consistent support.

Color



The Color Principle focuses on changing colors to enhance functionality or performance.

Remembering: What role does color play according to TRIZ principles?

Answer: Color plays a significant role in product design by influencing aesthetics, user perception, and functionality. It can also be used to convey information or enhance brand identity.

Understanding: Explain how color changes can influence user perception and functionality of products.

Answer: Color changes can affect user emotions and preferences; for example, bright colors may attract attention while muted tones may convey sophistication. Additionally, color coding can enhance usability by indicating functions (e.g., red for stop).

Color



Applying: Provide an example where color was strategically used to enhance product appeal or performance.

Answer: In consumer electronics, using vibrant colors for smartphone cases attracts younger audiences while also offering personalization options that enhance market appeal.

Analyzing: Analyze the psychological effects of color choices on consumer behavior.

Answer: Colors evoke specific emotions; for instance, blue is often associated with trust and reliability, making it a popular choice for financial institutions. Understanding these effects helps marketers tailor their strategies effectively.

Evaluating: Assess the impact of color on brand identity and recognition.

Answer: Consistent use of specific colors in branding enhances recognition; brands like Coca-Cola (red) and PepsiCo. (blue) leverage color to create strong associations with their identities.

Recycle & Recover



Remembering: What do recycling and recovering mean within TRIZ?

Answer: Recycling refers to processing used materials into new products to prevent waste, while recovering involves reclaiming valuable resources from waste streams for reuse.

Understanding: Describe methods by which recycling principles contribute to sustainable design practices.

Answer: Recycling principles encourage designing products for disassembly, using recyclable materials, and minimizing waste during production processes, promoting environmental sustainability.

Applying: Illustrate an example where recycling has been effectively integrated into product development.

Answer: Companies like Dell have implemented take-back programs that allow customers to return old electronics for recycling, reducing e-waste while recovering valuable materials for new products.

Recycle & Recover



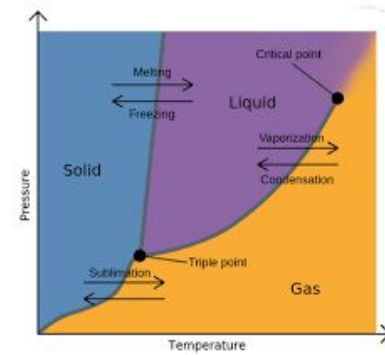
Analyzing: Analyze challenges faced when implementing recycling processes within manufacturing systems.

Answer: Challenges include sorting mixed materials efficiently, ensuring quality control during recycling processes, and balancing costs associated with recycling versus traditional disposal methods.

Evaluating: Evaluate economic benefits associated with recycling versus traditional disposal methods.

Answer: Recycling often reduces disposal costs and generates revenue from recovered materials. Long-term savings from reduced raw material purchases further enhance economic viability compared to traditional disposal methods.

Phase Transformation



Phase Transformation involves utilizing changes in physical states, such as volume or pressure, to solve problems innovatively.

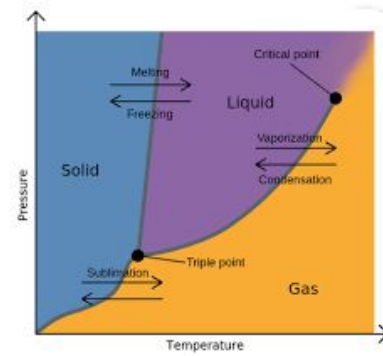
Remembering: What is phase transformation according to TRIZ principles?

Answer: Phase transformation refers to changes in the state of a material (e.g., solid to liquid) that can be utilized in engineering processes to achieve desired properties or functionalities.

Understanding: Explain its significance in material science and engineering applications.

Answer: Phase transformations are crucial in material science as they affect mechanical properties (e.g., strength), thermal behavior (e.g., melting points), and electrical conductivity of materials used in various engineering applications.

Phase Transformation



Applying: Provide examples where phase transformation has been utilized effectively.

Answer: Shape memory alloys (SMAs) undergo phase transformations that allow them to return to a predetermined shape when heated. This property is utilized in applications like stents and actuators.

Analyzing: Analyze how phase transformations affect material properties during manufacturing processes.

Answer: Phase transformations during processes like casting or forging can alter grain structure within metals, impacting strength and ductility. Understanding these effects is essential for optimizing manufacturing techniques.

Evaluating: Assess potential risks associated with phase transformations during production cycles.

Answer: Risks include unexpected changes in material properties leading to failures (e.g., brittleness) if phase transformations occur unintentionally during production due to improper temperature control.

Energy



The Energy Principle emphasizes the transformation and management of energy within systems to solve problems innovatively.

Remembering: What does energy refer to within TRIZ principles?

Answer: In TRIZ, energy refers to the capacity of a system or component to perform work, encompassing various forms such as mechanical, thermal, electrical, and chemical energy relevant to engineering problems.

Understanding: Describe different forms of energy relevant to engineering problems.

Answer: Common forms of energy in engineering include kinetic energy (energy of motion), potential energy (stored energy), thermal energy (heat), electrical energy (flow of electrons), and chemical energy (stored in molecular bonds).

Energy



Applying: Illustrate examples where energy efficiency has been improved through innovative designs.

Answer: Energy-efficient appliances, such as refrigerators with inverter technology, adjust power consumption based on load requirements instead of running at full capacity continuously, significantly reducing overall energy usage.

Analyzing: Analyze energy consumption patterns across various industries using TRIZ methodologies.

Answer: Industries like manufacturing have adopted TRIZ methodologies to identify inefficiencies; implementing automation and real-time monitoring systems has led to significant reductions in energy consumption during production processes.

Evaluating: Assess strategies for optimizing energy usage within existing systems.

Answer: Strategies include upgrading equipment to high-efficiency models, implementing smart controls that adjust usage based on demand fluctuations, and conducting regular energy audits to identify areas for improvement.

Imaging

The Imaging Principle involves visualizing solutions and understanding the relationships between system components.



Remembering: What role does imaging play according to TRIZ principles?

Answer: Imaging refers to the visual representation of information or objects used across various fields such as medicine (e.g., MRI scans), security (e.g., surveillance cameras), and entertainment (e.g., film production).

Understanding: Explain its importance across different fields such as medicine, security, or entertainment.

Answer: Imaging technologies enable non-invasive diagnostics in medicine, enhance surveillance capabilities in security systems through real-time monitoring, and transform storytelling techniques in entertainment by providing visual effects that engage audiences.

Imaging



Applying: Provide examples where imaging technologies have transformed traditional practices.

Answer: MRI technology revolutionized medical diagnostics by allowing detailed internal body imaging without invasive procedures. Similarly, digital cameras transformed photography by enabling instantaneous capture and editing compared to traditional film-based methods.

Analyzing: Analyze challenges faced by imaging technologies regarding accuracy or efficiency improvements.

Answer: Challenges include balancing resolution versus scanning speed; higher resolutions may require longer processing times leading to delays in diagnostics. Additionally, ensuring compatibility among diverse imaging systems poses integration issues impacting overall effectiveness.

Evaluating: Assess implications arising from advancements made within imaging systems over time.

Answer: Advancements have led to enhanced diagnostic capabilities improving patient outcomes significantly while also raising ethical concerns regarding privacy and data security that must be addressed proactively.

Environment



The Environment Principle focuses on creating solutions that minimize environmental impact while maximizing product functionality.

Remembering: How is environment defined within TRIZ principles?

Answer: In TRIZ, the environment encompasses all external factors influencing a system's performance, including physical surroundings (e.g., climate conditions), social dynamics (e.g., cultural norms), and regulatory frameworks affecting design choices.

Understanding: Describe interactions between engineered systems and their surrounding environments.

Answer: Engineered systems interact dynamically with their environments; for instance, buildings must withstand environmental stresses such as wind and rain while minimizing energy consumption through effective insulation strategies

Environment



Applying: Provide examples showcasing environmentally friendly designs inspired by TRIZ methodologies.

Answer: Solar panels exemplify environmentally friendly designs that harness renewable energy sources while minimizing reliance on fossil fuels, contributing significantly to sustainability efforts globally.

Analyzing: Analyze impacts resulting from neglecting environmental considerations during development phases.

Answer: Neglecting environmental considerations can lead to resource depletion and pollution, negatively affecting ecosystems and biodiversity while hindering progress towards sustainable development goals.

Evaluating: Assess long-term sustainability implications associated with various engineering practices concerning environmental factors.

Answer: Engineering practices must prioritize sustainability by ensuring minimal negative impacts on ecosystems; this includes adopting circular economy principles that promote recycling and resource efficiency throughout product lifecycles

Composition

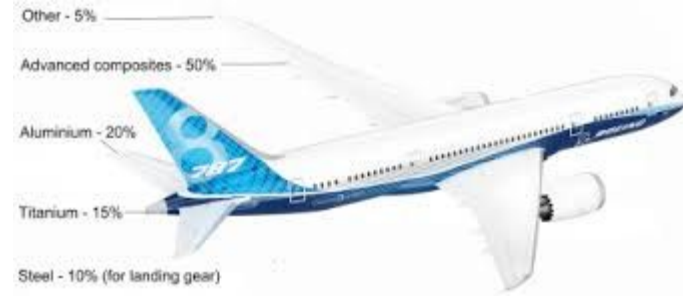
The Composition Principle refers to the idea of combining multiple elements or functions to create a more effective solution.

Remembering: What does composition refer to within TRIZ?

Answer: Composition refers to the combination of multiple components or materials together to create new functionalities or properties that achieve desired outcomes in engineering designs.

Understanding: Explain its significance when designing materials or systems from multiple components together effectively.

Answer: Composition allows engineers and designers to leverage diverse properties from different materials or components to optimize performance characteristics tailored to specific requirements.



Composition

Applying: Illustrate examples showcasing successful compositions

leading towards desired outcomes across industries (e.g., composites).

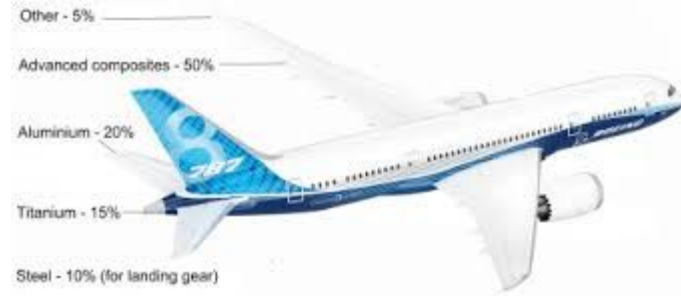
Answer: Composite materials like carbon fiber reinforced polymers combine lightweight fibers with resin matrices, resulting in strong yet lightweight structures widely used in aerospace and automotive applications.

Analyzing: Analyze challenges encountered while composing complex structures involving diverse elements together harmoniously (e.g., compatibility).

Answer: Challenges arise from ensuring compatibility among diverse elements; mismatches in thermal expansion coefficients can lead to structural failures necessitating thorough testing and validation during the design phase.

Evaluating: Assess trade-offs involved when choosing compositions based on performance requirements versus cost constraints (e.g., lightweight vs durability).

Answer: Trade-offs often require balancing performance against cost; lightweight compositions may reduce manufacturing costs but compromise durability, necessitating careful evaluation of material choices based on application needs



Economical

Remembering: How is economics defined concerning TRIZ principles?

Answer: Economics in TRIZ refers to the study of how resources are allocated efficiently within engineering processes, focusing on cost-effectiveness and value creation throughout product development.

Understanding: Describe factors influencing economic decisions throughout engineering processes.

Answer: Factors include initial investment costs, operating expenses, maintenance costs, potential savings from efficiencies gained through innovative designs, and the overall lifecycle costs associated with a product.

Applying: Provide examples illustrating successful economical strategies implemented within projects yielding significant savings.

Answer: Lean manufacturing techniques focus on minimizing waste while maximizing productivity; companies like Toyota have successfully implemented these strategies resulting in reduced costs and improved operational efficiency.



Economical



Analyzing: Analyze economic impacts resulting from adopting specific technologies over others based on lifecycle costs (e.g., initial investment vs maintenance).

Answer: Analyzing lifecycle costs reveals that while some technologies may have higher upfront costs, they can lead to lower maintenance expenses over time. For example, investing in high-efficiency HVAC systems may incur higher initial costs but result in significant long-term savings on energy bills.

Evaluating: Assess potential risks associated with ignoring economic considerations during decision-making phases impacting overall project viability (e.g., budget overruns).

Answer: Ignoring economic considerations can lead to budget overruns and project delays. For instance, failing to conduct a thorough cost analysis may result in unforeseen expenses that jeopardize project completion or profitability.

Surface Response

The Surface Response principle focuses on changing the properties or behaviors of a system over time to resolve conflicts between different parameters.



Remembering: What does surface response mean according to TRIZ principles?

Answer: Surface response refers to how the properties of a material's surface affect its performance characteristics, such as friction, wear resistance, and adhesion.

Understanding: Explain its relevance concerning material properties affecting performance characteristics (e.g., friction).

Answer: Surface response is critical because it determines how materials interact with their environment, influencing factors like durability and efficiency in applications such as bearings or coatings.

Surface Response



Applying: Illustrate examples showcasing successful surface treatments enhancing durability/longevity (e.g., coatings).

Answer: Hard anodizing aluminum enhances its surface hardness and corrosion resistance, significantly increasing the lifespan of components used in harsh environments like aerospace.

Analyzing: Analyze challenges faced while optimizing surface responses under varying conditions impacting reliability (e.g., wear resistance).

Answer: Challenges include balancing surface roughness for optimal friction versus wear resistance; too smooth surfaces may reduce friction but increase wear under load, necessitating careful design considerations.

Evaluating: Assess implications arising from neglecting surface responses during design phases leading towards failure modes (e.g., corrosion).

Answer: Neglecting surface responses can lead to premature failure due to issues like corrosion or fatigue; for example, untreated steel exposed to moisture can corrode quickly, compromising structural integrity.

Equipotential

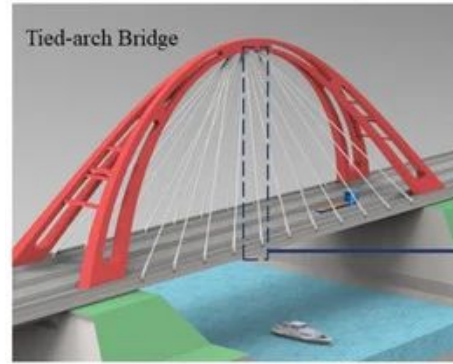
Remembering:

How is equipotential defined concerning TRIZ principles?

Answer: Equipotential refers to a condition in which all points within a system have the same potential energy, ensuring uniform distribution of forces and preventing imbalances.

Understanding: Describe its significance when designing systems ensuring uniformity across components avoiding imbalances (e.g., load distribution).

Answer: Equipotential design is crucial for stability; it prevents uneven load distribution that could lead to structural failures in systems like bridges or support structures.



Equipotential

Applying: Provide examples illustrating successful implementations achieving equipotentiality enhancing stability/reliability across structures (e.g., bridges).

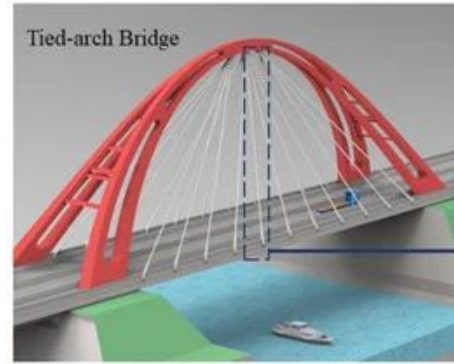
Answer: In bridge design, using tension cables that distribute loads evenly across the structure ensures equipotentiality, enhancing stability and preventing sagging or failure.

Analyzing: Analyze challenges encountered while maintaining equipotentiality under dynamic conditions affecting operational efficiency (e.g., vibrations).

Answer: Maintaining equipotentiality can be challenging in dynamic systems where vibrations cause shifting loads; designing damping systems is essential to counteract these effects and maintain stability.

Evaluating: Assess potential risks associated with ignoring equipotential considerations leading towards catastrophic failures across critical infrastructures (e.g., dams).

Answer: Ignoring equipotential considerations can lead to catastrophic failures; for instance, an imbalance in water pressure on a dam could cause structural failure, resulting in significant safety hazards and economic losses.



Static & Dynamic

The Static and Dynamic principle involves making fixed parts of a system movable or vice versa to optimize performance and adaptability



Remembering: How are static and dynamic defined concerning TRIZ principles?

Answer: Static refers to forces or systems that are at rest or in equilibrium, while dynamic refers to forces or systems that are in motion or changing over time.

Understanding: Describe interactions between static/dynamic forces influencing system behavior during operations affecting safety/performance characteristics (e.g., vibrations).

Answer: Static forces provide stability, while dynamic forces can introduce changes that affect performance; for example, vibrations from moving machinery can lead to fatigue in static components if not properly managed.

Static & Dynamic

Applying: Provide examples illustrating successful designs accommodating both static and dynamic loads ensuring stability/reliability across applications (e.g., vehicles).

Answer: Suspension systems in vehicles are designed to accommodate both static loads (weight of the vehicle) and dynamic loads (road conditions), ensuring a smooth ride while maintaining stability and safety.

Analyzing: Can you provide an example of applying the Static and Dynamic principle in a real-world scenario?

Answer: An example is designing an adjustable chair where the fixed components (like height) are made movable, allowing users to customize their seating position for comfort, thereby improving usability

Evaluating: Assess implications arising from neglecting static/dynamic interactions leading towards failure modes compromising overall integrity across critical infrastructures (e.g., bridges).

Answer: Neglecting static/dynamic interactions can result in structural failures; for example, a bridge designed only for static loads may collapse under dynamic loads from heavy traffic if not adequately reinforced.



Continuous & Intermittent



The Continuous and Intermittent principle refers to the use of continuous actions versus periodic (intermittent) actions in problem-solving and innovation

Remembering: How are continuous and intermittent defined concerning TRIZ principles?

Answer: Continuous refers to processes that operate without interruption, while intermittent refers to processes that operate in cycles with periods of activity followed by rest.

Understanding: Describe interactions between continuous/intermittent operations influencing system behavior during processes affecting productivity/efficiency (e.g., assembly lines).

Answer: Continuous operations tend to maximize throughput but may require more energy, whereas intermittent operations allow for maintenance but can lead to inefficiencies during downtime if not managed properly.

Continuous & Intermittent



Applying: Provide examples illustrating successful implementations accommodating both continuous/intermittent operations ensuring reliability across applications (e.g., power generation).

Answer: Hybrid power systems combine continuous sources like solar panels with intermittent sources like wind turbines, allowing for stable energy supply while accommodating fluctuations in generation capacity.

Analyzing: Analyze challenges faced while transitioning between continuous/intermittent modes impacting operational efficiency across diverse scenarios (e.g., machinery).

Answer: Transitioning can cause delays and inefficiencies; for instance, switching from continuous production to batch processing may require recalibrating machines, leading to downtime that affects overall productivity.

Evaluating: Assess implications arising from neglecting continuous/intermittent interactions leading towards failure modes compromising overall integrity across critical infrastructures (e.g., power plants).

Answer: Neglecting these interactions can result in failures; for example, a power plant designed primarily for continuous operation may face overheating issues if intermittent maintenance cycles are not adequately planned.

Servo Systems



A servo system can be analyzed as a hierarchical structure where each component contributes to achieving the desired function while minimizing contradictions and maximizing efficiency.

Remembering: How are servo systems defined concerning TRIZ principles?

Answer: Servo systems are automated control systems that use feedback loops to maintain desired output levels by adjusting input parameters based on real-time performance data.

Understanding: Describe interactions between servo systems influencing precision control during operations affecting safety/performance characteristics (e.g., robotics).

Answer: Servo systems provide precise control over position and speed; they adjust movements based on feedback signals from sensors, which enhances operational safety by preventing overshooting or erratic movements in robotics applications.

Servo Systems



Applying: Provide examples illustrating successful designs utilizing servo mechanisms ensuring reliability across applications (e.g., drones).

Answer: Drones use servo motors for precise control of flight surfaces like elevators, allowing for stable flight even under varying wind conditions by continuously adjusting based on sensor feedback.

Analyzing: Analyze challenges faced while optimizing servo responses under varying conditions impacting operational efficiency across diverse scenarios (e.g., temperature changes).

Answer: Servo systems may experience performance degradation due to temperature changes affecting motor efficiency; designing adaptive control algorithms helps maintain performance despite environmental variations.

Evaluating: Assess implications arising from neglecting servo interactions leading toward failure modes compromising overall integrity across critical infrastructures (e.g., automated factories).

Answer: Neglecting proper calibration and maintenance of servo systems can lead to failures in automated factories; misalignment due to worn components could result in product defects or equipment damage, impacting production quality.

Smart Systems



The Smart Systems principle focuses on the integration of intelligent systems that enhance efficiency, adaptability, and functionality.

Remembering: What defines a smart system within TRIZ principles?

Answer: A smart system is characterized by its ability to collect data through sensors, process information using algorithms or artificial intelligence, and make autonomous decisions based on real-time inputs to enhance functionality.

Understanding: Explain how smart systems integrate technology with user needs effectively.

Answer: Smart systems leverage advanced technologies such as the Internet of Things (IoT) to provide seamless interaction between devices and users, allowing for personalized experiences and improved efficiency in various applications, such as smart homes and automated manufacturing.

Smart Systems



Applying: Provide examples showcasing successful implementations of smart technologies enhancing functionality.

Answer: Smart thermostats, like the Nest, learn user behavior and adjust heating or cooling schedules automatically, optimizing energy consumption while maintaining comfort levels in homes.

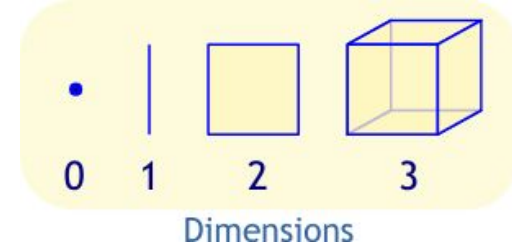
Analyzing: Analyze challenges faced when integrating smart technologies into existing infrastructures.

Answer: Challenges include compatibility issues with legacy systems, cybersecurity risks associated with increased connectivity, and the need for robust data management strategies to ensure privacy and security.

Evaluating: Assess implications arising from advancements made within smart technologies over time.

Answer: Advancements in smart technologies have led to increased automation and efficiency in various sectors but also raise ethical concerns regarding data privacy and the potential for job displacement due to automation.

Dimensions



The Dimensions Principle refers to the inventive principle that involves changing the dimensionality of an object or system. This principle encourages thinking beyond traditional linear approaches to problem-solving, promoting the exploration of two- or three-dimensional solutions.

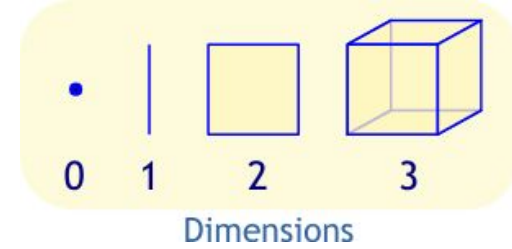
Remembering: How are dimensions defined concerning TRIZ principles?

Answer: Dimensions refer to measurable extents of physical objects, including length, width, height, and volume, which influence design choices and the feasibility of fitting components together within a system.

Understanding: Describe interactions between dimensions influencing spatial arrangements during operations affecting safety/performance characteristics.

Answer: Dimensions impact how components fit together in assemblies; for instance, insufficient clearance between moving parts can lead to mechanical failures or safety hazards during operation.

Dimensions



Applying: Provide examples illustrating successful designs accommodating dimensional constraints ensuring reliability across applications.

Answer: Modular furniture designs accommodate dimensional constraints by allowing users to customize configurations based on available space while ensuring structural integrity and ease of use.

Analyzing: Analyze challenges faced while optimizing dimensional configurations under varying conditions impacting operational efficiency across diverse scenarios.

Answer: Challenges include balancing compactness with accessibility; for example, designing equipment that is both space-efficient and easy to operate can require careful consideration of user ergonomics and operational workflows.

Evaluating: Assess trade-offs involved when choosing dimensions based on performance requirements versus cost constraints.

Answer: Trade-offs often involve balancing performance against manufacturing costs; larger dimensions may improve functionality but increase material costs and complexity in production processes, necessitating careful evaluation of design choices.